

Sauban Alam

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Summary

Third Year B.Tech student at IIT Jodhpur skilled in C++, Python, Data Structures, and Machine Learning. Passionate about software development, AI, and problem-solving with hands-on experience in algorithmic optimization, statistical analysis, and technical leadership.

Education

Indian Institute of Technology Jodhpur Jul 2024 – May 2028
Bachelor of Technology
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Statistical Analysis, Thermodynamics

Technical Skills

- **Programming Languages:** C++, Python, JavaScript, R
- **Web Development:** HTML, CSS, Node.js
- **Machine Learning & Data Analysis:** Regression, Supervised Learning, ANOVA, Hypothesis Testing
- **Data Structures & Algorithms:** Graphs, Trees, Hashing, STL, Greedy Algorithms, Dynamic Programming
- **Tools & Platforms:** Git, GitHub, RStudio, LaTeX

Projects

- Network Packet Routing Simulator — C++** 2026
- Developed a graph-based network simulator to model routing between interconnected nodes.
 - Implemented Dijkstra's, Prim's, and Kruskal's algorithms for optimized routing and network cost reduction.
 - Improved edge-processing efficiency using Union-Find (Disjoint Set) data structures.
 - Benchmarked shortest-path and MST-based routing approaches on large graph datasets.
- DSA Toolkit & Algorithm Visualizer — C++** 2026
- Built implementations of AVL Trees, BFS/DFS traversal, Hash Tables, and Disjoint Sets.
 - Solved algorithmic problems involving divide-and-conquer and greedy paradigms.
 - Created modular and reusable implementations for core data structure operations.
- Interactive Cognitive Performance Analysis Platform — R, HTML** 2026
- Designed and conducted a statistical study analyzing effects of noise and task complexity on arithmetic performance.
 - Performed ANOVA and regression analysis in R on experimental datasets.
 - Built a browser-based arithmetic testing application for automated participant data collection.

Leadership & Positions of Responsibility

- Assistant Head – Public Relations — IGNUS, IIT Jodhpur** Jan 2026 – Feb 2026
- Led outreach and publicity campaigns for IIT Jodhpur's cultural fest.
 - Coordinated communication with multiple teams to maximize student participation.
- Assistant Head – Publicity — Prometeo, IIT Jodhpur** Dec 2025 – Jan 2026
- Managed publicity operations and event promotions for the technical fest.
 - Collaborated with organizing teams to improve audience engagement.
- Assistant Head – Public Relations — Varchas, IIT Jodhpur** Oct 2025 – Nov 2025

- Coordinated participant communication and event management activities.
- Assisted in organizing large-scale student engagement initiatives.

Achievements

- Strong foundation in Data Structures, Algorithms, and Machine Learning.
- Active contributor to technical and organizational activities at IIT Jodhpur.
- Experienced in collaborative team environments and leadership roles.